

Research Idea: J133457.27+340238.7, J132035.40+340821.7,
J132443.68+323225.0, J130125.07+284038.0, J133944.15+274635.2,
J134308.83+302015.8, J131101.71+293442.0, J132051.75+312159.8,
J131313.45+335903.9, J130713.20+280249.0, J130636.39+275222.6,
J132838.16+333618.3, J131047.64+294235.6, J133306.57+330903.7,
J133947.34+300725.7, J130329.08+263301.7, J130801.65+264555.3,
J134017.95+262058.3, J085547.66+004739.4, J090655.44-000152.7

ALMA Thesis Planner (automated pipeline)

Title: A uniform CO(2–1) atlas, total-flux catalog, and rotation-amplitude measurement for the 20 ACA-mapped JINGLE spirals — with a calibrated molecular-disk size floor (ALMA 2018.1.00473.S)

Framing. Two co-equal legs. **Leg A (galaxy science):** a homogeneous catalog of total CO(2–1) fluxes, systemic velocities, integrated line widths, and — for the subset with resolved velocity gradients — a projected rotation amplitude from the moment-1 field, for all 20 targets. **Leg B (measurement calibration):** a *measured* answer to “how small a molecular disk can these ACA-only data size, and how large a one can they hold,” delivered as scale lengths for the galaxies that clear the floor and honest upper limits for those that don’t. The size–luminosity relation is the third-order product, reported for whatever subset survives, with the expectation stated up front that the sample — the 20 *brightest* JINGLE targets in JCMT CO(2–1) — is selection-compressed in CO luminosity and will yield a handful of real sizes among mostly limits.

Data and scope. Everything is public in the ALMA archive under project code ‘2018.1.00473.S’ (or per member OUS UID): pipeline-calibrated measurement sets plus QA2 image products, re-imageable in CASA. No new observations, no external catalogs, no cross-survey matching. The proposal’s own headline goals — resolved Kennicutt–Schmidt and gas-to-dust — need far-IR/UV/SFR maps this project does not supply and are excluded rather than half-attempted.

Line and frequency provenance. The targeted line is CO J=2–1, which the data description quotes at a **rest** frequency of 230.538 GHz. The tabulated **observed** windows span 220.4–227.7 GHz across the sample (~2 GHz per target). The description’s LLM annotation *infers* from those band centers a redshift range of roughly $z \approx 0.013$ – 0.047 — e.g. $z \approx 0.040$ for J132443.68+323225.0 from its 220.7–222.7 GHz tuning, $z \approx 0.017$ for J132051.75+312159.8 from its 225.7–227.7 GHz tuning. Those redshifts are assumed, back-derived from tuning rather than measured, and the annotation’s line list is explicitly flagged as generated from general knowledge and requiring verification against Splatalogue before imaging. Task one of the thesis is to verify the CO(2–1) rest frequency in Splatalogue and to re-derive each target’s systemic velocity from the line centroid of its own integrated spectrum before adopting any redshift or distance for angular-to-physical conversion.

Work plan (one semester).

1. **Inventory and atlas from the QA2 products.** Download all 20 OUSs. The archive QA2 cubes are the primary imaging product — no blanket re-imaging. Tabulate beams (4.9–6.7"), channel widths (~5.1–5.3 km/s), and the 5 mosaics vs. 15 single pointings. Apply primary-beam correction, then one common smooth-and-mask recipe for moment-0/moment-1 and an integrated spectrum per target. Re-image a target in ‘tclean’ only if its QA2 image fails a stated check (residual artifacts, missing pb correction), documented case by case.

2. **Leg A: fluxes, velocities, and rotation.** Total CO(2–1) flux from an aperture grown until the growth curve plateaus, with the growth curve itself as the flux uncertainty; systemic velocity and W50/W20 from the integrated spectrum. For each galaxy whose moment-1 map shows a monotonic gradient across ≥ 3 independent beams, measure a projected rotation amplitude from position–velocity cuts along the kinematic major axis. These are results about galaxies, not about the array, and they exist for the full sample regardless of how the size fitting goes.

3. **Leg B: size fitting, in the image plane, forward-modeled.** Fit an **inclined exponential disk** — free parameters: central surface brightness, scale length, axis ratio, position angle, centroid — to each primary-beam-corrected moment-0 map by convolving the model with that target’s restoring beam and minimizing residuals in the image plane. This is chosen deliberately over ‘uvmodelfit’, which offers no exponential component, assumes a single phase center (so it cannot handle the 5 mosaicked targets X1bd2, X1be6, X1bea, X1bf2, X1c1e), and fits per channel rather than the line-integrated emission. Image-plane forward-modeling handles mosaics and single pointings identically and operates on a single velocity-collapsed map; its cost is that spatial filtering is *not* handled natively, which is exactly what step 4 measures.

4. **Calibration: uv-plane injection, image-plane recovery.** Inject synthetic inclined exponential disks of known scale length, axis ratio, and flux into copies of the real measurement sets (‘ft’/‘simobserve’) for **two** representative targets — one near the best beam/S/N, one near the worst — then image and re-fit them through the *identical* step-1/step-3 calls. Grid ~5 sizes $\times$ 3 fluxes $\times$ 2 inclinations $\times$ 2 targets. Outputs: the minimum recoverable scale length as a function of S/N and size/beam, the bias in recovered axis ratio, and the fractional flux lost as a function of source size relative to each target’s shortest-baseline limit. Sizes are quoted only above the resulting floor; everything below gets an upper limit.

5. **Largest recoverable scale.** From each measurement set’s own shortest-baseline distribution, compute a per-target maximum recoverable scale and flag any galaxy whose fitted size approaches it. State explicitly that these are 7-m-array-only data with no total-power product listed, and that the JCMT single-dish CO(2–1) fluxes that selected the sample are not part of this data description — so the usual external flux-recovery check is unavailable and this internal bound substitutes for it.

On inclinations — stated once, plainly. The step-3 fit is *inclined*: it returns axis ratio and position angle as free parameters, so the resolved subset does get photometric inclinations, and the scale lengths for that subset are deprojected. Two consequences are carried honestly. (i) For galaxies below the size floor, the axis ratio is unconstrained; their scale-length limits are quoted as face-on-equivalent and labeled as such. (ii) Rotation amplitudes and line widths are reported **twice** — as observed, inclination-uncorrected values for all 20, and additionally deprojected using the fitted axis ratio *only* for galaxies where step 4 shows that axis ratio is recovered without significant bias. No CO Tully–Fisher calibration is claimed from ~a handful of deprojected points; the deprojected values are published as a catalog column for future use.

Explicitly out of scope: any fixed-physical-radius concentration index (at these beams a 3 kpc aperture is comparable to the beam over much of the sample), gas-to-dust and Kennicutt–Schmidt work requiring external maps, and any Tully–Fisher fit.

Optional extension if time remains: stack line-free channels across all 20 targets for an average Band 6 continuum constraint, given per-target continuum sensitivities of $\sim 0.34\text{--}0.66$ mJy/beam that likely leave most individual galaxies undetected. Interpretation limit to be stated if attempted: because the targets sit at different inferred redshifts, such a stack averages *different rest-frame wavelengths* around ~ 1.2 mm across a heterogeneous sample, so it constrains a sample-mean flux, not a well-defined single-wavelength dust measurement.

Deliverables: a documented, reproducible analysis repository; an atlas of moment-0/moment-1 maps, position–velocity cuts, and integrated spectra for all 20 targets; a machine-readable catalog of total fluxes with growth-curve uncertainties, verified systemic velocities, observed and (where justified) deprojected line widths and rotation amplitudes, exponential scale lengths or limits with fitted axis ratios, and per-target largest-recoverable-scale bounds; the two-target injection–recovery calibration of the size and axis-ratio floors; and the size–luminosity figure, color-coded by inferred redshift so the distance-dependent resolution bias is visible rather than hidden.